Fundamental Postulate of Statistical Mechanics:

For an isolated system, in equilibrium all accesible microstates with the same energy are equally probable. That is,

$$P(i)\Big|_{\mathcal{E}_i}^{N} = \frac{1}{N}$$

N → N total microstates with energy $\mathcal{E}_i$

$P(i)$ → Probability of state i

$\mathcal{E}_i$ → at energy $\mathcal{E}_i$

This is the "microcanonical" ensemble, or (N, V, E)

Hypothesis: Entropy (S) is some unknown function ϕ of the number of states that a system can adopt (W).

Consider two non-interacting states A and B.

we say then that $S_A = \phi(W_A)$

$S_B = \phi(W_B)$

now combine the two states:

$$S_{A+B} = \phi(W_{A+B})$$

and: $W_{A+B} = W_A W_B$

but: $S_{A+B} = S_A + S_B$ because entropy is extensive.

$$\therefore \phi(W_A W_B) = \phi(W_A) + \phi(W_B).$$

What function has this property?

$$\boxed{S\Big|_{N,V,E} = k_B \ln\left(W(N,V,E)\right)}$$